

Ella Lewis

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TECHNICAL SKILLS

Languages: Python, C/C++, Verilog-A, VHDL, Assembly

Design & Simulation: Altium, KiCAD, PSPICE/LTspice, PLECS, MATLAB, Keysight IC-CAP, Microsoft Office

Characterization: Oscilloscope, double pulse, wafer-level probing, semiconductor analyzer, power analyzer, FLIR thermal imaging, logic analyzer, e-Load, SMU

Protocol: RS-485, UART, CAN, I2C, USB

PUBLICATIONS

- **E. Lewis**, R. Fang, D. Xu, J. Alant, R. Griffin, L. Wei, “RF GaN HEMT Compact Modelling with High-Temperature Effects for Extreme Environment Applications,” *21st Canadian Semiconductor Science and Technology Conference (CSSTC)*, August 2026. Oral presentation.

EXPERIENCE

Automotive Product Applications Engineering Intern

Aug 2025 – Aug 2026

Infinion Technologies Inc. | *PSPICE, PLECS, Altium, MATLAB* | 8 month co-op

Ottawa, ON

- **Designed the end to end** of a DSP daughter card PCB in Altium for a 900V GaN bidirectional FETs, achieving 1.04ns precision PWM with programmable deadtime, frequency, duty and **60% gate-source noise reduction**.
 - * **Developed pulsed technique** for >200 kHz-switching FET characterization by adding on-command long pulse to mitigate measurement error from clamping diode transients for accurate $R_{ds(on)}$ measurement.
 - * **Built a unified evaluation board test suite**, consolidating AC-ZVS, DC-ZVS, buck/boost, and dynamic $R_{ds(on)}$ pulsing into one interface; implemented nanosecond-precise control C firmware (FSM, ISR).
 - * **Electrically validated** in AC/DC test suite up to 650V 35A and **pushed design internally**, replacing 4 separate test setups and increasing reliability, facilitating customer live demos and field app trainings.
- **Brought up and debugged** two 650V GaN BDS and integrated power-stage EVBs in half-bridge, bench-testing up to 500V/100A across -50°C to 150°C with double-pulse, power analyzer, and thermal imaging; defined device performance limits, device modeling and customer requirement qualification.
- **Calibrated PSPICE and PLECS models** to device switching and thermal data and built a buck converter simulation template, delivering **loss-simulation accuracy within 96%** for customer converter design-in.

Undergraduate Research Assistant (GaN RF e-HEMT)

Jan 2026 – Present

University of Waterloo | *Keysight IC-CAP, Verilog-A, MATLAB*

Waterloo, ON

- Extended the MVSG model for high temperature 76mm GaN RF e-HEMTs, fitting mobility, V_{th} , s-parameters and gate-leakage trends to measured on-wafer IV data from 22°C to 500°C for high-temperature RF circuit design.
- Performed **wafer-level probing and electrical characterization** to acquire device data used for bidirectional device compact-model development.

Electrical Distribution Intern

Jan 2025 – May 2025

EQUNAS (Plan-Group) | *Python, AutoCAD, Revit, Bluebeam*

Vancouver, BC

- Built a **Python-based QA/QC automation tool** to continuously audit LV and MV board designs against spec sheets, improving detection of non-compliant components to **95% accuracy**, enabling project milestone tracking.
- Validated **70% of 600V–1.2kV MV** substation infrastructure by cross-referencing short-circuit and arc flash studies, enabling detection of **30+** transformer impedance and breaker miscoordination issues.

PROJECTS

[View Portfolio](#)

Injector Sensor Hub - Waterloo Rocketry | *KiCad, C, PCB Design*

May 2025

- Designed and laid out a 2-layer CAN sensor board for a liquid rocket engine, conditioning 6 pressure-transducer and Hall flow channels into an MCU ADC and reporting to the main control unit over CAN.
- Flew on Polaris, **which reached 63,497 ft** in Aug 2026, the highest amateur liquid rocket launch on record.

USB-C PD Power Bank | *KiCad, USB Power Delivery, STM32G0*

June 2026 – Present

- Designed 4-layer dual-port USB-C PD power bank around a 1S Li-ion cell with dual PD, and a I2C buck-boost stage negotiating PDOs over CC for both charge-in and 20W source-out.
- Sized the power budget against cell droop to hold the input current limit; integrated charging, OTG source-out, fuel gauge, and NTC cell-temperature sensing under STM32 supervision over I2C.

EDUCATION

University of Waterloo

Waterloo, ON

B.Eng. Honours Electrical Engineering (Co-op), President's Scholarship, 3.95 GPA

Sept 2024 – Present